1. Cover Page

This section explains Cover Page. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Cover Page. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Cover Page. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

2. Certificate

This section explains Certificate. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Certificate. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Certificate. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

3. Acknowledgement

This section explains Acknowledgement. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Acknowledgement. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Acknowledgement. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

4. Abstract

This section explains Abstract. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Abstract. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Abstract. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

5. Introduction

This section explains Introduction. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Introduction. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Introduction. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

6. Problem Statement

This section explains Problem Statement. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Problem Statement. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Problem Statement. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

7. Objectives

This section explains Objectives. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Objectives. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Objectives. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

8. Scope

This section explains Scope. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Scope. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Scope. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

9. Literature Review

This section explains Literature Review. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Literature Review. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Literature Review. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

10. Machine Learning

This section explains Machine Learning. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Machine Learning. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Machine Learning. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

11. Deep Learning

This section explains Deep Learning. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Deep Learning. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Deep Learning. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

12. CNN

This section explains CNN. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains CNN. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains CNN. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

13. FaceNet

This section explains FaceNet. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains FaceNet. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains FaceNet. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

14. DeepFace

This section explains DeepFace. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains DeepFace. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains DeepFace. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

15. Dataset

This section explains Dataset. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Dataset. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Dataset. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

16. Data Preprocessing

This section explains Data Preprocessing. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Data Preprocessing. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Data Preprocessing. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

17. Face Detection

This section explains Face Detection. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Face Detection. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Face Detection. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

18. Feature Extraction

This section explains Feature Extraction. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Feature Extraction. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Feature Extraction. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

19. Face Matching

This section explains Face Matching. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Face Matching. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Face Matching. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

20. System Architecture

This section explains System Architecture. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains System Architecture. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains System Architecture. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

21. Methodology

This section explains Methodology. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Methodology. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Methodology. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

22. Flowchart

This section explains Flowchart. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Flowchart. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Flowchart. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

23. Training

This section explains Training. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Training. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Training. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

24. Testing

This section explains Testing. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Testing. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Testing. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

25. Evaluation Metrics

This section explains Evaluation Metrics. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Evaluation Metrics. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Evaluation Metrics. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

26. Confusion Matrix

This section explains Confusion Matrix. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Confusion Matrix. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Confusion Matrix. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

27. Attendance Module

This section explains Attendance Module. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Attendance Module. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Attendance Module. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

28. Database Design

This section explains Database Design. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Database Design. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Database Design. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

29. Implementation

This section explains Implementation. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Implementation. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Implementation. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

30. Python Libraries

This section explains Python Libraries. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Python Libraries. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Python Libraries. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

31. Results

This section explains Results. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Results. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Results. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

32. Advantages

This section explains Advantages. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Advantages. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Advantages. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

33. Limitations

This section explains Limitations. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Limitations. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Limitations. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

34. Future Scope

This section explains Future Scope. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Future Scope. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Future Scope. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

35. Applications

This section explains Applications. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Applications. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Applications. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

36. Conclusion

This section explains Conclusion. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Conclusion. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains Conclusion. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.

37. References

This section explains References. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains References. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration. This section explains References. The project proposes an AI-based face recognition attendance management system using Machine Learning and Deep Learning. Images are collected, preprocessed, faces detected with OpenCV/MTCNN, embeddings extracted using CNN-based FaceNet/DeepFace, matched against enrolled users, and attendance stored in a database. The report discusses algorithms, workflow, implementation, evaluation metrics such as accuracy, precision, recall and F1-score, advantages, limitations, and future improvements including liveness detection, cloud deployment and mobile integration.